

Worksheet 3A: Quadratics — FOIL and Expanding — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. $t^2 + 7t + 10$
2. $p^2 + 9p + 18$
3. $m^2 + 11m + 28$
4. $r^2 + 10r + 9$
5. $k^2 + 10k + 16$
6. $n^2 + 10n + 25$
7. $w^2 + 6w + 9$
8. $s^2 + 11s + 10$
9. $h^2 + 12h + 36$
10. $u^2 + 8u + 16$
11. Result is $x^2 + 2ax + a^2$ (perfect square trinomial)
12. $2ax$

Part B

13. (a) 49 (b) 25 (c) No
14. (a) 49 (b) 29 (c) 20
15. $a^2 + 2ab + b^2$
16. The $2ab$ term (the middle term)
17. (a) 24 (b) 49 (c) Yes
18. (a) 9 (b) 21 (c) No (d) $a^2 - 2ab + b^2$
19. (a) 27 (b) 9 (c) No
20. multiplication; addition (or subtraction)

Part C

- 21. $t^2 + 2t - 15$
- 22. $p^2 + 2p - 8$
- 23. $m^2 - 5m - 6$
- 24. $r^2 - 7r + 10$
- 25. $k^2 - 10k + 21$
- 26. $n^2 - 8n + 16$
- 27. $w^2 - 16w + 64$
- 28. $s^2 - 25$
- 29. $h^2 - 81$
- 30. $u^2 - 49$
- 31. Middle terms cancel; result is $x^2 - a^2$
- 32. $x^2 - a^2$

Part D

- 33. $2t^2 + 7t + 3$
- 34. $3p^2 + 14p + 8$
- 35. $2m^2 + 7m - 15$
- 36. $4r^2 - 7r - 2$
- 37. $6k^2 - k - 2$
- 38. $10n^2 - 11n - 6$
- 39. $4w^2 + 12w + 9$
- 40. $9s^2 - 6s + 1$
- 41. $16h^2 - 25$
- 42. $4u^2 - 49$

Part E

- 43. $t^2 + 16t + 64$
- 44. $p^2 - 10p + 25$
- 45. $m^2 - 121$
- 46. $9r^2 + 24r + 16$
- 47. $25k^2 - 20k + 4$

48. $36n^2 - 1$
49. $4w^2 + 36w + 81$
50. $16s^2 - 9$

Part F

51. 1
52. 1, 1
53. $a^2 + 2ab + b^2$; Coefficients: 1, 2, 1
54. $a^3 + 3a^2b + 3ab^2 + b^3$; Coefficients: 1, 3, 3, 1
55. $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$; Coefficients: 1, 4, 6, 4, 1
56. 1, 4, 6, 4, 1
57. Each number is the sum of the two numbers directly above it
58. 1, 5, 10, 10, 5, 1
59. Pascal's Triangle
60. $a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$
61. AB, AC, AD, BC, BD, CD (6 combinations)
62. 10 ways (row 5, position 3 is 10)

Part G

63. $t^3 + 5t^2 + 10t + 12$
64. $p^3 + 2p^2 - 7p - 2$
65. $m^3 + 1$ (sum of cubes!)
66. $r^3 - 1$ (difference of cubes!)
67. $2k^3 - 5k^2 + k + 2$
68. $n^3 + 6n^2 + 11n + 6$
69. $w^3 - 3w^2 + 3w - 1$
70. $8t^3 + 12t^2 + 6t + 1$
